Chapter 1

The Black–Scholes–Merton Formula: Derivations

Chapter 15 states the formula and explains what its terms mean. This note carries out the integral that produces it, shows the equivalence with solving the differential equation, and derives the dividend and American-approximation results the chapter uses.

1.1 The Risk-Neutral Integral

Under the risk-neutral measure, [TN14] established that

$$\ln S_T \sim N\big(m,\, s^2\big)\,, \qquad m = \ln S_0 + \Big(r - \tfrac{\sigma^2}{2}\Big)\,T, \qquad s = \sigma\sqrt{T} \tag{TN15.1}$$

and the call is the discounted expected payoff,

$$c = e^{-rT}\,\mathbb{E}^Q[\max(S_T - K,\,0)] = e^{-rT}\,\mathbb{E}^Q[(S_T - K)\mathbf{1}\{S_T > K\}] \tag{TN15.2}$$

Split it into two pieces:

$$c = e^{-rT}\,\underbrace{\mathbb{E}^Q[S_T\mathbf{1}\{S_T > K\}]}_{A} - e^{-rT}K\,\underbrace{Q(S_T > K)}_{B} \tag{TN15.3}$$

The second piece: B

Write $\ln S_T = m + sZ$ with Z standard normal. Then $S_T > K$ if and only if

$$Z > \frac{\ln K - m}{s}$$

so, using the symmetry $1 - N(x) = N(-x)$,

$$B = N\!\left(\frac{m - \ln K}{s}\right) = N\!\left(\frac{\ln(S_0/K) + \big(r - \sigma^2/2\big)\,T}{\sigma\sqrt{T}}\right) = N(d_2) \tag{TN15.4}$$

which identifies d_2 and confirms Chapter 15's statement that $N(d_2)$ is the risk-neutral probability of exercise.

The first piece: A

Here the useful step is to complete the square. Writing $S_T = e^{m+sZ}$ and ϕ for the standard normal density,

$$A = \int_{z^*}^{\infty} e^{m+sz}\,\phi(z)\,dz = \frac{1}{\sqrt{2\pi}}\int_{z^*}^{\infty} \exp\!\Big(m + sz - \tfrac{z^2}{2}\Big)\,dz, \qquad z^* = \frac{\ln K - m}{s}$$

The exponent is

$$m + sz - \frac{z^2}{2} = m + \frac{s^2}{2} - \frac{(z-s)^2}{2}$$

so, substituting $y = z - s$,

$$A = e^{m+s^2/2}\,\frac{1}{\sqrt{2\pi}}\int_{z^*-s}^{\infty} e^{-y^2/2}\,dy = e^{m+s^2/2}\,N(s - z^*) \tag{TN15.5}$$

Two substitutions finish it. First, $e^{m+s^2/2} = S_0 e^{rT}$—this is the lognormal moment result of [TN14], and it is the same cancellation of $\sigma^2/2$ noted there. Second,

$$s - z^* = \frac{m - \ln K}{s} + s = d_2 + \sigma\sqrt{T} = d_1$$

so $A = S_0 e^{rT} N(d_1)$.

Assembling

Substituting A and B into (TN15.3):

$$\boxed{c = S_0 N(d_1) - K e^{-rT} N(d_2)} \tag{TN15.6}$$

The put follows from put–call parity rather than a second integral, since parity holds identically.

Note where each term came from. $N(d_2)$ arose as a probability; $N(d_1)$ arose from completing the square, that is, from the shift $z \mapsto z - s$ induced by weighting each state by S_T. That shift is the whole content of Chapter 15's remark that $N(d_1)$ is not a probability—it is the same probability computed under a measure in which the stock, rather than cash, is the numeraire. Chapter 30 develops that idea in general.

1.2 Equivalence with the Differential Equation

Chapter 15 offers two routes to (TN15.6). They agree, and the reason is worth stating.

Define $V(S,t) = e^{-r(T-t)}\mathbb{E}_t^{Q}[h(S_T)]$ for any payoff h. By the tower property of conditional expectation, the discounted value $e^{-rt}V$ is a Q-martingale. Applying Itô's lemma to $e^{-rt}V(S,t)$ under the risk-neutral dynamics $dS = rS\,dt + \sigma S\,dW^{Q}$:

$$d\big(e^{-rt}V\big) = e^{-rt}\left(V_t + rSV_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS} - rV\right)dt + e^{-rt}\sigma S V_S\,dW^{Q}$$

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A martingale has zero drift. Setting the dt coefficient to zero gives

$$V_t + rSV_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS} = rV \tag{TN15.7}$$

which is equation (15.1). *The differential equation and the risk-neutral expectation are the same statement*: one says the discounted price has no drift, the other computes the expectation that follows. This equivalence is the Feynman–Kac relation, and it is why Chapter 20 can solve pricing problems either by simulating expectations or by solving partial differential equations.

1.3 Dividends

A continuous yield. With a yield q, the risk-neutral drift becomes $r - q$: the asset's total return must be r, and q of it arrives as income, so the price grows at $r - q$. Repeating Section 1.1 with $m = \ln S_0 + (r - q - \sigma^2/2)T$ changes (TN15.5) to $A = S_0 e^{(r-q)T} N(d_1)$ and leaves (TN15.4) in the same form, giving

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(S_0/K) + \left(r - q + \sigma^2/2\right) T}{\sigma\sqrt{T}} \tag{TN15.8}$$

Known cash dividends, and the volatility subtlety. Chapter 15 subtracts the present value of dividends and applies the formula to $S^* = S - \mathrm{PV}(D)$. The justification is that S^* is the risky component: S equals a certain part, $\mathrm{PV}(D)$, plus an uncertain part which is assumed lognormal.

The subtlety the chapter flags can be made precise. If S^* has volatility σ^* and $S = S^* + \mathrm{PV}(D)$, then the volatilities of the two are related by

$$\sigma^* \approx \sigma \, \frac{S}{S^*} = \sigma \left(1 - \frac{\mathrm{PV}(D)}{S}\right)^{-1} \tag{TN15.9}$$

because the same absolute risk is carried by a smaller base. For a stock yielding 6% with a two-year option, $\mathrm{PV}(D)/S \approx 0.113$, so σ^* exceeds the observed σ by roughly 13%—not negligible. Using the unadjusted volatility understates the option.

1.4 Black's Approximation

Chapter 15 states the rule; here is why it takes that form.

Section 11.9 established that an American call on a stock paying a single dividend at t_1 can be optimally exercised only at two moments: immediately before t_1, or at expiry T. Exercise elsewhere is dominated, because the interest saved by deferring the strike payment exceeds any benefit until the dividend is imminent.

The value of an American call is the maximum over admissible exercise policies. Restricting attention to those two:

$$C \approx \max\left(\underbrace{c(S_0 - \mathrm{PV}(D),\, K,\, T)}_{\text{hold to } T},\ \underbrace{c(S_0,\, K,\, t_1)}_{\text{exercise at } t_1} \right) \tag{TN15.10}$$

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The first term treats the option as European to T on the dividend-adjusted price. The second treats it as European to t_1 with no adjustment, since a holder exercising then captures the dividend and the adjustment does not apply.

The approximation *understates*, and the reason is now visible: it evaluates a fixed choice between two policies made today, whereas the true American holder decides at t_1 with knowledge of S_{t_1}. Being able to choose after observing is worth something, so

$$C \geq \max(\cdot, \cdot)$$

with equality only in the degenerate cases where one policy dominates in every state. The error is largest when the two terms are close, which is exactly when the exercise decision is genuinely uncertain.

Further Reading

Black and Scholes (1973) derive the formula by solving the differential equation; Merton (1973) uses the continuously rebalanced portfolio and also handles dividends and the American call. Cox and Ross (1976) is where the risk-neutral integral of Section 1.1 first appears as a general method. For the Feynman–Kac equivalence of Section 1.2 at a readable level, Baxter and Rennie is the shortest route; Shreve's second volume gives the careful version.

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